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1 Goodness of fit tests and categorical data analysis

We are often interested in determining whether or not a particular probabilistic model is appropriate for a given random phenomenon. This determination often reduces to testing whether a given random sample comes from some specified, or partially specified, probability distribution. For example, we may a priori feel that the number of industrial accidents occurring daily at a particular plant should constitute a random sample from a Poisson distribution. This hypothesis can then be tested by observing the number of accidents over a sequence of days and then testing whether it is reasonable to suppose that the underlying distribution is Poisson. Statistical tests that determine whether a given probabilistic mechanism is appropriate are called goodness of fit tests. The classical approach to obtaining a goodness of fit test of a null hypothesis that a sample has a specified probability distribution is to partition the possible values of the random variables into a finite number of regions. The numbers of the sample values that fall within each region are then determined and compared with the theoretical expected numbers under the specified probability distribution, and when they are significantly different the null hypothesis is rejected.

A goodness-of-fit test, in general, refers to measuring how well do the observed data correspond to the fitted (assumed) model. Like we will see in linear regression, in essence, the goodness-of-fit test compares the observed values to the expected (fitted or predicted) values. A goodness-of-fit statistic tests the following hypothesis:

$$H_0 : \text{The model } M_0 \text{ fits}$$

vs

$$H_1 : \text{The model } M_0 \text{ does not fit or, some other model } M_1 \text{ does}$$

Most often, the alternative hypothesis represents the saturated model which perfectly fits the data because each observation has a separate parameter. But this M_1 model could be other than the saturated one.

1.1 Goodness of fit test when all parameters are specified

Let us now consider the simplest example of the goodness-of-fit test with categorical data illustrated with an example. In this particular case, the objective is testing particular proportion values for a categorical variable. Assume X is a categorical variable that represents the result obtained when rolling a dice. X takes six possible values (categories) and, if the dice is fair, the probability of each value (category) is equal to $1/6$. If we roll the dice once, $X \sim Multin(1, \mathbf{p})$ where $\mathbf{p}$ is a vector of six entries, all of them equal to $1/6$. We want to test the hypothesis that there is an equal probability of six faces by comparing the observed frequencies to those expected under the assumed model. To do that, we roll the dice 30 times and observe the frequency of each side of the dice. Here, the number of categories, k is equal to six. Then:

$$\begin{cases} H_0 & : & X \sim Multin(30, \mathbf{p}) \\ H_1 & : & \text{any of the components in } \mathbf{p} \neq 1/6 \end{cases}$$

There are two test statistics available for this test: the Pearson goodness-of-fit test statistic and the deviance statistic (or likelihood ratio test). The first one compares the observed counts with the counts expected under the null hypothesis in the following way:

$$T = \sum_{j=1}^k \frac{(O_j - E_j)^2}{E_j}$$

where $O_j = X_j$ is the observed count in category j , and $E_j = np_j$ is the expected count in category j under the assumption that the null hypothesis is true. T measures how closely the model, in this case $Multin(30, \mathbf{p})$, "fits" the observed data. It has an approximate chi-square distribution with $k - 1$ degrees of freedom when H_0 is true. This allows us to use the chi-square distribution to find critical values and p-values for establishing statistical significance.

- If the sample proportions $\hat{p}_j$ (i.e., saturated model) are exactly equal to the model's p_j for categories $J = 1, \dots, k$, then $O_j = E_j$ for all j , and T will be zero. That is, the model fits perfectly.
- If the sample proportions $\hat{p}_j$ deviate from the p_j , then T is positive. Large values of T mean that the data do not agree well with the assumed/proposed model M_0 proposed in H_0 .

An old-fashioned rule of thumb is that the χ^2 approximation works well provided that n is large enough to have $E_i = np_i \geq 5, \forall i$. Nowadays, most agree that we can have $E_i < 5$ for some of the cells (say, 20% of them). Some of the E_i can be as small as 2, but none of them should fall below 1. It is also recommended that $O_i \geq 3, \forall i$ and the number of categories is ≥ 5 . Categories can be merged to form larger ones, increasing the number of observed and expected values and reducing the number of categories.

Example Imagine we are trying to determine whether or not a dice is fair rolling it 30 times. We get the following frequencies for the face which ends up on top, displayed in what it is called a one-way table, that is, a table containing the observed frequencies of one categorical variable:

Value	Count
1	3
2	7
3	5
4	10
5	2
6	3

Under the null hypothesis that the dice is fair, $X \sim Multin(30, \mathbf{p})$ where $p_i = 1/6, i = 1, \dots, 6$. This is our assumed model, and under this H_0 , the expected counts are $E_i = 30/6 = 5$ for each cell. The value for the statistic $T = 9.2$. Under the null hypothesis, T follows a χ^2_5 and the corresponding p-value would be:

$$P(\chi^2_5 \geq 9.2) = 0.1$$

Given this p-value, with the usual 5% significance level, we fail to reject the null hypothesis. We simply don't have enough evidence to conclude it's false. That is, the fair-dice model doesn't fit the data exactly, but the fit isn't bad enough to conclude that the dice is unfair, given our significance threshold of 0.05.

Example: Tomato Phenotypes Tall cut-leaf tomatoes were crossed with dwarf potato-leaf tomatoes, and $n = 1611$ offspring were classified by their phenotypes.

Phenotypes	Count
tall cut-leaf	926
tall potato-leaf	288
dwarf cut-leaf	293
dwarf potato-leaf	104

Genetic theory says that the four phenotypes should occur with relative frequencies $9 : 3 : 3 : 1$, and thus are not all equally as likely to be observed. The dwarf potato-leaf is less likely to be observed than the others. Do the observed data support this theory?

Under the null hypothesis, the probabilities for the four categories of tomato are:

$$p_1 = \frac{9}{16}, p_2 = p_3 = \frac{3}{16}, p_4 = \frac{1}{16}$$

and the expected frequencies under the proposed model are:

$$E_i = np_i, E = c(906.2, 302.1, 302.1, 100.7)$$

We calculate the statistic and find that $T = 1.47$. The p-value based on the χ^2_3 distribution is approximately equal to 0.69. Therefore, we fail to reject the null hypothesis and accept (by default) that the data are consistent with the genetic theory.

Another example A contractor who purchases a large number of fluorescent lightbulbs has been told by the manufacturer that these bulbs are not of uniform quality but rather have been produced in such a way that each bulb produced will, independently, either be of quality level A, B, C, D, or E, with respective probabilities .15, .25, .35, .20, .05. However, the contractor feels that he is receiving too many type E (the lowest quality) bulbs, and so he decides to test the producer's claim by taking the time and expense to ascertain the quality of 30 such bulbs. Suppose that he discovers that of the 30 bulbs, 3 are of quality level A, 6 are of quality level B, 9 are of quality level C, 7 are of quality level D, and 5 are of quality level E. Do these data, at the 5 percent level of significance, enable the contractor to reject the producer's claim?

We can use R to find the p-value for the test:

```
chisq.test(c(3,6,9,7,5),p=c(0.15,0.25,0.35,0.2,0.05))
```

As some expected values fall below 5 there is a warning using this test. We can switch to using simulation instead adding the options (`simulate.p.value = T`, `B=10000`).

Thus the hypothesis would not be rejected at the 5 percent level of significance (but since it would be rejected at all significance levels above .053, the contractor should certainly remain skeptical).

1.2 Goodness of fit test: unspecified parameters

We can also perform goodness of fit tests of a null hypothesis that does not completely specify the values of the parameters for the distributions under the null. In this case, these parameters have to be estimated by the method of maximum likelihood. If there are m of these parameters and if the sample size n is high, the test statistic T will have, when H_0 is true, approximately a chi-square distribution with $k - 1 - m$ degrees of freedom. In other words, one degree of freedom is lost for each parameter that needs to be estimated from the data.

1.3 Test of independence for contingency tables

Here, we will consider problems in which each member of a population can be classified according to two distinct characteristics which shall be denoted by X-characteristic and Y-characteristic. We suppose that there are r possible values for the X characteristic and s possible values for the Y characteristic. Let p_{ij} be the joint probability that a randomly chosen member of the population will have X-characteristic i and Y-characteristic j , for $i = 1, \dots, r$ and $j = 1, \dots, s$. Let $p_{i\cdot}$ denote the probability that an arbitrary member of the population will have X-characteristic i , and $p_{\cdot j}$ is the probability it will have Y-characteristic j . We are interested in testing the hypothesis that a population member's X- and Y -characteristics are independent, that is,

$$H_0 : p_{ij} = p_{i\cdot}p_{\cdot j}, \forall i = 1, \dots, r \quad \text{and} \quad j = 1, \dots, s$$

versus

$$H_1 : p_{ij} \neq p_{i\cdot}p_{\cdot j}, \text{ for at least one pair } (i, j) \quad i = 1, \dots, r \quad \text{and} \quad j = 1, \dots, s$$

Under the null hypothesis, and if n members of the population have been sampled, with a total of n_{i+} having the X-characteristic i and a total of n_{+j} having the Y-characteristic j , the number of observations expected to be jointly observed under the characteristic (i, j) is:

$$E_{ij} = np_{ij} = np_{i+}p_{+j} = \frac{n_{i+}n_{+j}}{n}$$

and we have observed O_{ij} of such observations. The more distant these two quantities are the more evidence we have against the null. The statistic used here will be the following. Under the null hypothesis, its distribution is approximately a χ^2 with $(r-1)(s-1)$ degrees of freedom:

$$T = \sum_{i=1}^r \sum_{j=1}^s \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi_{(r-1)(s-1)}^2$$

An old-fashioned rule of thumb is that the χ^2 approximation works well provided that n is large enough to have $E_{ij} = np_{ij} \geq 5$ for every i, j . Nowadays, most agree that we can have $E_{ij} < 5$ for some of the cells (say, 20% of them). Some of the E_{ij} can be as small as 2, but none of them should fall below 1. If this happens, then the χ^2 approximation isn't appropriate, and the test results are not reliable. Anyway, permutation tests can be easily design (R will provide functions to run them directly) to obtain simulated p-values that overcome this inconvenience.

1.3.1 Computing the degrees of freedom for this test

To compute the expected values under the null hypothesis, we have to estimate $r-1 + s-1$ parameters; $r-1$ values for p_{i+} and $s-1$ values for p'_{+j} since $\sum_{i=1}^r p_{i+} = 1$ and $\sum_{j=1}^s p_{+j} = 1$. Hence, we actually need estimate $r+s-2$ parameters, and since each population member has $k = rs$ different possible values, it follows that the resulting test statistic will, for large n , have approximately a chi-square distribution with $rs - 1 - (r + s - 2) = (r-1)(s-1)$ degrees of freedom.

An Example: Political Affiliation A sample of 300 people was randomly chosen, and the sampled individuals were classified as to their gender and political affiliation: Democrat, Republican, or Independent. The following contingency table displays the data obtained:

X	Y			Total
	Democrat	Republican	Independent	
Women	68	56	32	156
Men	52	72	20	144
Total	120	128	52	300

Thus, for instance, the contingency table indicates that the sample of size 300 contained 68 women who classified themselves as Democrats, 56 women who classified themselves as Republicans, and 32 women who classified themselves as Independents. Use these data to test the hypothesis that a randomly chosen individual's gender and political affiliation are independent.

Let's name $X = \text{gender}$ with values 1 = Women and 2 = Men and $Y = \text{Political Affiliation}$ with values 1 = Democrat, 2 = Republican and 3 = Independent. The table above gives the values for

O_{ij} , $i = 1, 2$ and $j = 1, 2, 3$. To obtain the expected values under the hypothesis of independence of the two characteristics we proceed as follows:

$$E_{11} = \frac{n_{1+}n_{+1}}{n} = \frac{156 \cdot 120}{300} = 62.4, \quad E_{12} = \frac{156 \cdot 128}{300} = 66.56, \text{ etc.}$$

To compute the test statistic:

$$T = \frac{(68 - 62.4)^2}{62.4} + \frac{(56 - 66.56)^2}{66.56} + \dots + \frac{(20 - 24.96)^2}{24.96} = 6.433$$

and we compute the p-value as follows:

$$p - \text{value} = P(\chi_2^2 \geq 6.433) = 0.04$$

To obtain the p-value in R, just type `pchisq(6.433, 2, lower.tail=FALSE)`.

Using a 5% significance level, we reject the null hypothesis that gender and political affiliation of members of the population are independent.

Important observation For the example above about political affiliation, another way in which we could gather data is to fix in advance the numbers of men and women in the sample and then choose random samples of those sizes from the subpopulations of men and women. That is, rather than let the numbers of women and men in the sample be determined by chance, we might decide these numbers in advance. Because doing so would result in fixed specified values for the total numbers of men and women in the sample, the resulting contingency table is often said to have fixed margins (since the totals are given in the margins of the table). When the data are collected in this manner, the same hypothesis test can still be used to test for the independence of the two characteristics.

An Example of fixed margins: A randomly chosen group of 20,000 nonsmokers and one of 10,000 smokers were followed over a 10-year period. The following data relate the numbers of them that developed lung cancer during that period.

	Smokers	Nonsmokers	Total
Lung cancer	62	14	76
No lung cancer	9938	19986	29924
Total	10000	20000	30000

Test the hypothesis that smoking and lung cancer are independent. Use the 1% level of significance.

If smoking and lung cancer were independent, the estimates of the expected number to fall in each ij cell are:

$$E_{11} = \frac{n_{1+}n_{+1}}{n} = \frac{76 \cdot 10000}{30000} = 25.33, \quad E_{12} = \frac{76 \cdot 10000}{30000} = 60.67$$

$$E_{21} = \frac{29924 \cdot 10000}{30000} = 9974.67, \quad E_{22} = 19949.33$$

The value of the test statistic is $T = 79.83$. To compute the p-value we use a χ^2_1 . The p-value is 0. We reject the null hypothesis that whether a randomly chosen person develops lung cancer is independent of whether that person is a smoker.

Note that this is an example and it is not *real data*. This kind of studies are actually done using a **retrospective setting**. Find out more later on in this document.

An Example: Coronary Heart Disease This is an example of a 2×4 contingency table. In this study, $n = 1329$ patients were classified by serum cholesterol level (mg/100 cc) and whether they had been diagnosed with coronary heart disease (CHD).

X: CHD status	Y: Cholesterol level				Total
	0-199	200-219	220-259	260+	
CHD	12	8	31	41	92
No CHD	307	246	439	245	1237
Total	319	254	470	286	1329

One variable is binary and has two outcomes. The other variable has four levels - this is most likely a continuous variable, but it has been grouped into four intervals. The kind of questions we answer performing an independence test are: Is having coronary heart disease independent of the cholesterol level in ones body? Is there any evidence of a relationship/association between cholesterol level and heart disease?

To solve the problem by hand, we need to compute the value of the χ^2 statistic. First, we need to compute the expected values under the independence model:

$$E_{11} = \frac{n_{1+}n_{+1}}{n} = \frac{92 \cdot 319}{1329} = 22.082, \quad E_{12} = \frac{92 \cdot 254}{1329} = 17.583, \text{ etc.}$$

Then, the value of the statistic is $T = 35.028$. Under the null hypothesis $T \sim \chi^2_3$. The corresponding p-value is:

$$P(\chi^2_3 \geq 35.028) = 1.2 \times 10^{-7}$$

We reject the null hypothesis of independence because of the big value of the chi-square statistic. Notice the degrees of freedom are equal to $(4 - 1) \cdot (2 - 1) = 3$, and thus the p-value is very low. Therefore, through the test for independence, we have demonstrated beyond a reasonable doubt that a relationship exists between cholesterol and CHD.

A good statistical analysis, however, should not end with the rejection of a null hypothesis. Once we have demonstrated that a relationship exists, we may be interested in which cells were particularly revealing. To do this we consider computing and evaluating the residuals.

The following code enters the data in R and gets the p-value for the test. You can take advantage of the function `chisq.test()` to obtain the expected values (under the null hypothesis) and the value for the residuals. Those cells with the highest residuals are responsible for the rejection of the null hypothesis.

```

heart <-c(12,8,31,41,307,246,439,245)
heart<-matrix(heart,4,2)
heart=t(heart)
result<-chisq.test(heart)
result
result$observed
result$expected
result$residuals
result$stdres

```

Recall that residuals tell how far off are the expected and observed values for each cell, under the assumed model. They tell us which cells drive the lack of fit. We can check for Pearson and standardized residuals calculated under the null model. The Pearson residual for a cell in a two-way table is:

$$r_{ij} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}}$$

where the chi-squared statistics is then $T = \sum_i \sum_j r_{ij}^2$. The standardized (adjusted) Pearson residual for a cell in a two-way table is

$$r_{ijs} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}(1 - p_{i+})(1 - p_{+j})}}$$

A standardized Pearson residual has an approximate $N(0, 1)$ distribution. A value that exceeds 2 or 3 in absolute value, therefore, suggests a lack of fit. For the heart disease example data, the residual in the cell (1, 4) is

$$r_{14s} = \frac{41 - 19.79}{\sqrt{19.79(1 - \frac{92}{1239})(1 - \frac{268}{1239})}} = 5.575$$

and would suggest lack of fit of the independence model.

1.3.2 Chi-Squared test is also used as a test of Homogeneity

A test of independence is based on data on two categorical variables from a single random sample. In a test of homogeneity, data are observed on a single categorical variable from random samples selected independently from two or more populations or from an experiment where subjects are assigned at random to different treatment groups with a categorical response. The chi-squared statistic for a test of homogeneity has the same form as the chi-squared statistic for a test of independence. The differences in the two tests are in the nature of the hypotheses being tested and the study design employed for collecting data. Computationally, the chi-square test statistic and P-value are identical for the two tests. For simplicity, we will refer to the test statistic as a chi-squared test of independence for both study designs.

Example: Lawsuits in Surgery To determine if a malpractice lawsuit is more likely to follow certain types of surgeries, random samples of three different types of surgeries were studied, and the following data resulted.

Type of Operation	Number Sampled	Number leading to a lawsuit
Heart Surgery	400	16
Brain Surgery	300	19
Appendectomy	300	7

Test the hypothesis that the percentages of the surgical operations that lead to lawsuits are the same for each of the three types.

If we define a success (Yes) as the surgery that leads to a lawsuit, we can arrange the data in the following contingency table and test if the proportions of successes is the same in the three groups defined by the three different types of surgery:

Type	Outcome	Yes	No	Total
	Heart	16	384	400
	Brain	19	281	300
	App.	7	293	300
	Total	42	958	1000

The value for the test statistic is $T = 5.526$ and the p-value, computed using a χ^2_2 is:

$$P(\chi^2_2 \geq 5.526) = 0.0631$$

At a 5% level of significance, we fail to reject the null hypothesis and we don't have enough evidence to claim that the proportions are different in the three types of surgery.

1.3.3 Chi-Squared and the test comparing proportions in 2×2 tables

In practice, contingency tables of size 2×2 are very common. These occur in summarizing responses of two groups on a binary response variable, such as belief in life after death (yes, no).

Denote the population proportion of success by p_1 in group 1 and p_2 in group 2. Then $(1 - p_1)$ and $(1 - p_2)$ are the population proportions of failures. The table displays the notation. The rows are the groups to be compared and the columns are the response categories.

Population Proportions			
Groups	Success	Failure	Total
1	p_1	$1 - p_1$	1.0
2	p_2	$1 - p_2$	1.0

The above table with counts instead of proportions is the 2×2 contingency table:

Counts			
Groups	Success	Failure	Total
1	n_{11}	n_{12}	$n_1 = n_{11} + n_{12}$
2	n_{21}	n_{22}	$n_2 = n_{21} + n_{22}$

If the response variable is independent of the group, then $p_1 = p_2$, so the conditional distributions are identical. Hence, the null hypothesis $\mathbf{H}_0 : \text{independence}$ is, equivalently, the homogeneity hypothesis $\mathbf{H}_0 : \mathbf{p}_1 = \mathbf{p}_2$ of equality of population proportions. You have probably

studied a way to conduct this test as a z test (based on the Normal distribution). The test statistic is the difference between the sample proportions divided by its standard error, that is

$$Z = \frac{(\hat{p}_1 - \hat{p}_2)}{se_0}, \quad se_0 = \sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

with se_0 the standard error estimated under the presumption that H_0 is true. The estimates are:

$$\hat{p}_1 = \frac{n_{11}}{n_1} \quad \text{and} \quad \hat{p}_2 = \frac{n_{21}}{n_2}$$

The pooled estimate $\hat{p}$ is the proportion of successes for the whole group of observations:

$$\hat{p} = \frac{n_{11} + n_{21}}{n_1 + n_2}$$

In fact, using algebra, it can be shown that the chi-squared statistic for 2×2 tables is related to this Z statistic by

$$T = Z^2$$

In other words, square the Z statistic, and you get the T statistic. For $H_1 : p_1 \neq p_2$, the Z test has exactly the same p-value as the chi-squared test. So a two-sided test based on comparing the z-statistic to a $N(0, 1)$ distribution is identical to comparing T from the test of independence to a chi-squared distribution with χ_1^2 .

Important observation When the null hypothesis of independence is rejected, the nature of dependence—its direction and magnitude—needs to be measured. A statistic that measures the direction and magnitude of the relationship between two variables is called an effect size or a measure of association. One of the most intuitive measures of association is the difference in proportions, which compares the relative frequency of important characteristics between two groups. Even though the difference of two proportions is very easy to interpret, one problem with using $p_1 - p_2$ is that when *the success* is a rare event, the individual probabilities p_1 and p_2 are both small, i.e., close to zero. The absolute value of the difference will be close to zero even when the effect is strong. Consider for example the yearly risk of cancer death to be .005 in one population (1 death every 200 people) and .01 in another population (two deaths every 200 people); most health officials would consider this difference to be important (the second one is twice the first one), even though this difference is very small (their difference is 0.005). This suggests the use of two other common measures of association which compare the relative value of the proportions, rather than the absolute values: the relative risk *of a success* (risk ratio) and the odds ratio.

An Example: Vitamin C This is an example of a 2×2 contingency table. It illustrates the equivalence between the Chi-squared test for independence and the z-test (using a normal distribution) comparing proportions in 2×2 tables.

This is an example of a double-blind study investigating the therapeutic value of vitamin C (ascorbic acid) for treating common colds. The study was conducted during a 2 week period on a sample of 279 French skiers. There are two discrete variables each having two levels - hence the two-way table. The incidence of common colds in these French skiers is given in the following table:

Treatment	Cold		Total
	Cold	No Cold	
Placebo	31	109	140
Vitamin C	17	122	139

Each cell indicates levels of both traits. For example, 31 skiers were given a placebo and contracted cold while 109 did not. Here are some questions that we may ask regarding this data:

1. Is the probability that a member of the placebo group contracts a cold the same as the probability that a member of the ascorbic group contracts a cold?
2. Are the type of treatment and cold status associated or independent? Here, independence means that having a placebo or ascorbic acid has no relationship with having a cold or otherwise.
3. What are the odds of getting a cold for those taking ascorbic acid (vitamin C)?

To answer question 1 in a way learned in a basic statistics course, one can solve the following hypothesis test:

$$\left. \begin{array}{l} H_0 : p_1 = p_2 \\ H_1 : p_1 \neq p_2 \end{array} \right\}$$

where p_1 is the population proportion of skiers having a cold in the placebo group and p_2 is the same proportion but in the Vitamin C group. As stated above, this is done with a test based on the Normal distribution:

$$\hat{p}_1 = \frac{31}{140} = 0.22, \quad \hat{p}_2 = \frac{17}{139} = 0.12, \quad \hat{p} = \frac{31 + 17}{279}, \quad z = 2.19$$

and the corresponding p-value:

$$p - \text{value} = 2P(Z \geq 2.19) = 0.0285$$

Thus, we have evidence to reject the null hypothesis and conclude the two proportions are different and treatment and cold status are not independent. As we have shown above, to conduct a test of independence would be equivalent to what we have already done. In this case, we have to compute the expected values for each cell of the contingency table under the hypothesis of independence:

$$E_{11} = \frac{140 \cdot 48}{279} = 24.08, \dots, E_{22} = \frac{139 \cdot 231}{279} = 115.086$$

Next, we have to compute the statistic $T = 4.8114$ and the p-value as:

$$P(\chi_1^2 \geq 4.8114) = 0.0282$$

We can see that $T = 4.8114 \approx Z^2 = (2.19)^2 = 4.7961$. The hypothesis of independence is thus rejected.

We could use the difference between the conditional sample proportions $\hat{p}_1 - \hat{p}_2$ to try and measure the magnitude of the relationship between these two variables. In this example, the estimated or sample difference of proportions of getting a cold is 0.1. Is this difference *big* or *small*? Although it is difficult to quantify the difference, what we can do is compute a (large sample) confidence interval for such difference. As most elementary statistics books show, this 95% confidence interval is given by (assuming large samples and thus using a normal approximation):

$$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

Inserting our values,

$$0.1 \pm 1.96 \times 0.045, \quad 95\% \text{ Conf. Int} \quad (0.01, 0.19)$$

We are 95% confident that the true difference in proportions of people getting cold given the placebo or a vitamin C treatment is somewhere between 1% and 19%. Note that the value of 0 (corresponding to no difference) does not fall within that boundary. Since the interval is entirely positive, we could conclude that the proportion of colds in vitamin C group is smaller than in the placebo group.

Let's now use this Vitamin C example as a pretext to define two more common measures of association in 2×2 contingency tables. The first one, the **relative risk** compares the relative value of the conditional proportions rather than the absolute values. It is the ratio of the conditional probabilities of success under the two different conditions; i.e., the probability of success given row 1 and the probability of success given row 2. Defining the variables Z =cold status (1 =cold, 0 =no cold) and Y =treatment (1 =placebo, 2 =VitC), the definition is the following:

$$r = \frac{P(Z = 1|Y = 1)}{P(Z = 1|Y = 2)} = \frac{p_1}{p_2}$$

In this particular example, the estimate for r is:

$$\hat{r} = \frac{\frac{31}{140}}{\frac{17}{139}} = \frac{0.2214}{0.1223} = 1.81$$

The skiers who took a placebo are more likely (nearly two times more likely) to get a cold in comparison with the skiers who took Vitamin C.

Testing for $H_0 : r = 1$ is equivalent to testing for independence in a 2×2 table. For our current example, $r = 1.8$ and it has a 95% confidence interval of (1.0519, 3.1162). Thus, at 0.05 level, we can reject the independence model.

Keep in mind that the relative risk is a better measure of association than the difference in proportions when cell probabilities are close to 0 and 1, i.e., when they are in the tails of the

probability distribution. Relative risk has far more application in medical literature and epidemiology than difference of proportions.

The second measure of association is perhaps the most commonly used: the **odds ratio**. The odds are ratios of probabilities of “success” and “failure” for a given row, or a ratio of conditional probabilities of the same conditional distribution. The odds of getting a cold versus not getting a cold given that a person took placebo are:

$$odds_1 = \frac{P(Z = 1|Y = 1)}{P(Z = 0|Y = 1)} = \frac{p_1}{1 - p_1} = \frac{\frac{31}{140}}{\frac{109}{140}} = \frac{31}{109} = 0.2844$$

The second odds (given that ascorbic acid was taken) are:

$$odds_2 = \frac{P(Z = 1|Y = 2)}{P(Z = 0|Y = 2)} = \frac{p_2}{1 - p_2} = \frac{17}{122} = 0.1393$$

Properties of odds:

1. If odds equal to 1, “success” and “failure” are equally likely.
2. If odds > 1 , then “success” is more likely than “failure”.
3. If odds < 1 , then “success” is less likely than “failure”.
4. $\frac{1}{odds}$ are odds of a “failure” (if odds are odds of a “success”)

The **odds ratio**, is the ratio of $odds_1$ and $odds_2$ (or vice versa):

$$\theta = \frac{P(Z = 1|Y = 1)/P(Z = 0|Y = 1)}{P(Z = 1|Y = 2)/P(Z = 0|Y = 2)}$$

The natural estimate of θ is the sample crossproduct ratio:

$$\hat{\theta} = \frac{31 \times 122}{17 \times 109} = 2.04$$

In this example, $\hat{\theta} = 2.04$ means that:

1. The odds of getting a cold given a placebo are 2.04 times greater than the odds of getting a cold given vitamin C.
2. The odds of getting a cold given vitamin C are $1/2.04 = 0.49$ times the odds of getting a cold given a placebo.
3. Getting a cold is less likely given vitamin C than given a placebo.

Properties of the odds ratio:

If $\theta = 3$, the odds of “success” in row 1 are 3 times greater than the odds of success in row 2; individuals in row 1 are more likely to have a “success” than those in row 2. If $\theta = 0.3$, the odds of “success” in row 1 are 0.3 times the odds of the row 2; the odds of “success” in row 2

are $1/0.33 = 3.33$ times the odds in row 1.

The relationship between odds and probabilities can be expressed as:

$$odds_1 = \frac{p_1}{1 - p_1} \Leftrightarrow p_1 = \frac{odds_1}{1 + odds_1}$$

If the variables are independent, then $p_1 = p_2$, $odds_1 = odds_2$ and

$$\theta = \frac{odds_1}{odds_2} = 1$$

If the variables are not independent such that $p_1 > p_2$, then $odds_1 > odds_2$, and

$$1 < \theta$$

If the variables are not independent such that $p_1 < p_2$, then $odds_1 < odds_2$, and

$$0 < \theta < 1$$

If both p_1 and p_2 are small in the population, then the odds ratio and relative risk will be close since $\frac{1-p_1}{1-p_2}$ will be close to 1. The odds ratio θ does NOT depend on the marginal distribution of either variable. If the categories of both variables are interchanged, the value of θ does not change. If the categories of one variable are switched, the odds ratio in the new re-arranged table will equal $1/\theta$.

If we interchange the roles of Y and Z , i.e., Y is now the response variable and Z is the explanatory variable, we still get the same θ . So θ can be regarded as a function of the parameters of $P(Y|Z)$ as well. This is important in retrospective studies.

Finally, note that the sample odds ratio will equal zero or ∞ if any $n_{ij} = 0$. Some authors suggest adding $1/2$ to each cell count and then recalculating the sample odds ratio and its standard error to avoid this issue.

A note on prospective and retrospective studies Suppose variable Z represents a condition (disease, the response variable) that is relatively rare in a population (e.g. lymphoma), and we want to assess whether another characteristic or behavior Y (e.g. smoking, the explanatory variable) could be a risk factor for Z . The obvious way to study this is to follow a group of smokers ($Y = 1$) and a group of nonsmokers ($Y = 2$) over time, and see which ones eventually develop lymphoma ($Z = 1$) and which do not ($Z = 2$). This is called a **prospective study**. The exposed and unexposed groups are determined at the start of the study and both groups are disease-free. While it makes logical sense in determining a significant relationship, it can be very

- time-consuming (we have to wait for a long time for the problem condition to develop)
- inefficient (we may need very large samples to obtain enough subjects with $Z = 1$)

An alternative is the **retrospective study**, in which we first locate a group of subjects with lymphoma ($Z = 1$) and identify which are smokers and which are not. Here a diseased group is determined first and is retrospectively assessed for exposure status. Then we locate another group of subjects who are in some sense "comparable" but who do not have lymphoma ($Z = 2$) and identify which are smokers and which are not. In the **retrospective study**, we have "sampled on the outcome", choosing individuals on the basis of Z and then observing Y .

The interchangeability of Y and Z means that the usual roles of "explanatory" and "response" variables can be reversed, which could be extremely useful for research. Because **the odds ratio is invariant to exchanging Y and Z** , the odds ratio from a retrospective study should be about the same as the odds ratio from a prospective study in which we sampled individuals according to their Y values and collected information on Z . A retrospective study provides no information about the overall incidence of Z in the population because the proportions of cases with $Z = 1$ and $Z = 2$ were decided by the investigator. However, it does provide consistent estimates of the odds ratio indicating the effect of Y on Z .

Another example In a famous experiment to determine the efficacy of aspirin in preventing heart attacks, 22000 healthy middle-aged men were randomly divided into two equal groups, one of which was given a daily dose of aspirin and the other a placebo that looked and tasted identical to the aspirin. The experiment was halted at a time when 104 men in the aspirin group and 189 in the control group had had heart attacks. Use these data to test the hypothesis that the taking of aspirin does not change the probability of having a heart attack.

Using the data above, we fill in the following contingency table:

Treatment	Heart Attack		Total
	Yes	No	
Aspirin	104	10896	11000
Placebo	189	10811	11000

The code to perform a χ^2 test of independence, compute the odds ratio and a confidence interval for the odds ratio is:

```
library(vcd)
data_aspirin <- matrix(c(104,189,10896,10811),
                      nrow = 2,
                      dimnames = list("Treatment" = c("Aspirin", "Placebo"),
                                      "Heart Attack" = c("Yes", "No")))
chisq.test(data_aspirin)
or=oddsratio(data_aspirin, log=FALSE)
confint(or, log=FALSE)
```

The low p-value for the χ^2 test sheds evidence that the two variables are not independent. To measure the strength of the dependence, we compute odds ratio, $\hat{\theta} = 0.545$. The odds for having a heart attack in the treatment group are estimated to be half the odds for having a heart attack in the control group.

2 Two way tables: Dependent Data

When categorical responses are paired, such as when subjects respond to the same question at two different time points, we could carry out the usual test of independence for categorical variables, but more specifically, we may be more interested in whether proportions are equal or in the extent that the subjects' opinions agree over the two-time points.

Suppose you were asked shortly after his inauguration in 2021, "Do you think Joe Biden will be effective as President of the United States?" Then, after four years, suppose you are asked, "Do you think Joe Biden has been effective as President of the United States?" Whatever your responses are over the two time points, they will not be independent because they come from the same person. This is not the same situation as being chosen by chance in two random samples at those time points. If the same people are necessarily asked again four years later, then we really have only one sample with pairs of (dependent) responses. In dependent samples, each observation in one sample can be paired with an observation in the other sample. Sometimes, responses come for different individuals but they are related in some way: for instance, pairs of siblings. The same happens when the same sample of people is asked, for example: Do you believe in Heaven? and Do you believe in Hell? The two Yes-No answers are given by the same person and thus, the responses are dependent. When studying matched pairs data we might be interested in comparing the margins of the table, i.e., a row proportion versus a column proportion. In the last example, imagine we ask the same people twice and we get the following table:

	Belief in Hell		
Belief in Heaven	Yes	No	Total
Yes	955	162	1117
No	9	188	197
Total	964	350	1314

We might be interested in comparing the proportion of people who believe in Heaven (estimated from the data as $\hat{p}_1 = \frac{1117}{1314}$) to the proportion who believed in Hell (estimated from the data as $\hat{p}_2 = \frac{964}{1314}$).

The null hypothesis of no difference (marginal homogeneity) in a 2×2 table is $H_0 : p_1 = p_2$ and it is equivalent to the hypothesis that the off-diagonal counts are equal. Note that the diagonal elements are not important here. All the information required to determine whether there's a difference in beliefs is contained in the off-diagonal elements.

For general square $I \times I$ tables, the hypothesis of marginal homogeneity is different from the hypothesis of symmetry, and the latter is a stronger hypothesis; symmetry introduces more structure in the square table. In a 2×2 table, however, these two are the same test.

2.1 McNemar's test for 2×2 tables

This is the usual test of marginal homogeneity (and symmetry) for a 2×2 table. Let's label the four cells of a 2×2 contingency table as a, b, c and d .

a	b
c	d

The null hypothesis of marginal homogeneity in 2×2 tables is equivalent to the null $H_0 : b = c$, that is, that the off-diagonal counts are equal.

Suppose that we treat the total number of observations in the off-diagonal as fixed, $n^* = b + c$. Under the null hypothesis, each of b and c is assumed to follow a $\text{Binomial}(n^*, 0.5)$ distribution. The justification is that, under the null hypothesis, we have $b + c$ total *trials* that can either result in cell $(1, 2)$ or $(2, 1)$ with probability 0.5. And provided that n^* is sufficiently large, we can use the usual normal approximation to the binomial:

$$z = \frac{b - 0.5 * n^*}{\sqrt{0.5(1 - 0.5)n^*}} = \frac{1/2(b - c)}{\sqrt{0.5^2 n^*}} = \frac{b - c}{\sqrt{b + c}}$$

where $0.5n^*$ and $0.5(1 - 0.5)n^*$ are the expected value and variance for b under the null ($b \approx \text{Binomial}(n^*, 0.5)$). Under H_0 , z is approximately standard normal. This approximation works well provided that $n^* \geq 10$. The p-value would depend on the alternative hypothesis. If H_1 is that the proportion of people who believes in Heaven is greater than the proportion who believe in Hell, then the p-value would be:

$$P(Z \geq z), \quad Z \sim N(0, 1)$$

where z is the observed value of the statistic. A lower-tailed alternative would correspondingly use the lesser-than probability, and a two-sided alternative would double the one-sided probability. Alternatively, for a two-sided alternative we may use:

$$z^2 = \frac{(b - c)^2}{b + c} \underbrace{\sim}_{\text{under } H_0} \chi_1^2$$

When the sample size is small, we can compute exact probabilities (p-values) using the binomial probability distribution.

Example Suppose that we ask a group of $n = 1600$ voters whether they approve of the President's performance at two points in time. The two characteristics are approval of the President's performance (Presidential approval: approve & disapprove) at two periods of time (Time 1 and Time 2). It is only sensible to assume that these two variables should be related, because the same 1600 voters are being asked the same question at two points in time. An individual's political views at one time are likely to be somewhat similar to his or her views at another time. Has the president's approval rating changed over time? This is the same as asking if the margins of the table are the same, which can be done with the test of marginal homogeneity or with McNemar's test. Here, the latter is illustrated.

Time 1 -Time 2	Approved	Disapproved	Total
Approved	794	150	944
Disapproved	86	570	656
Total	880	720	1600

Entering the data in R and output of the test:

```
datos <- matrix(c(794, 86, 150, 570),  
               nrow = 2,  
               dimnames = list("Approved" = c("Yes", "No"),  
                              "Disapproved" = c("Yes", "No")))  
mcnemar.test(data, correct=F)
```

McNemar's Chi-squared test

data: data

McNemar's chi-squared = 17.356, df = 1, p-value = 3.099e-05

Given the low p-value, there is very strong evidence that the probability of approval at the time of the first survey was different (higher than) that the time of the second survey.